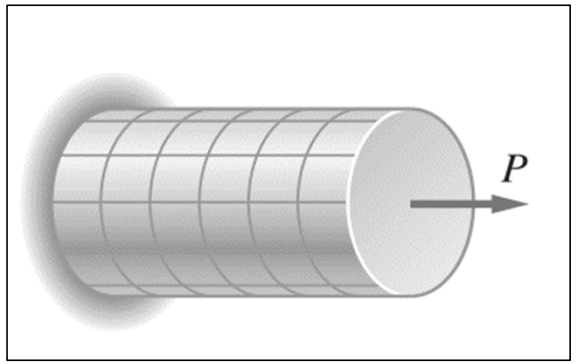
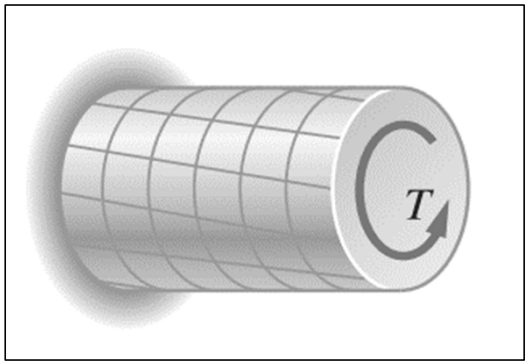
Transverse/Flexural/Bending Loading

Flexural/Bending Loads

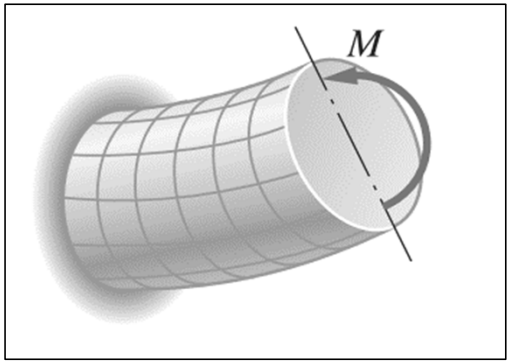
- Recall:



Axial Loading

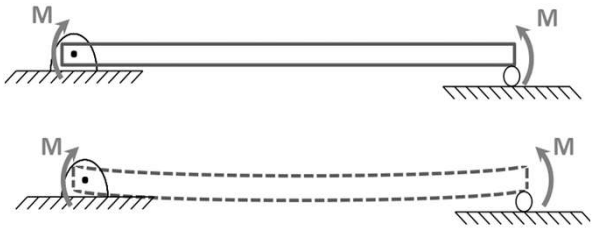


Torsional Loading

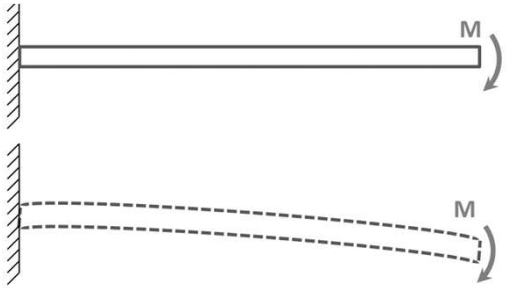


(Pure) Flexural Loading

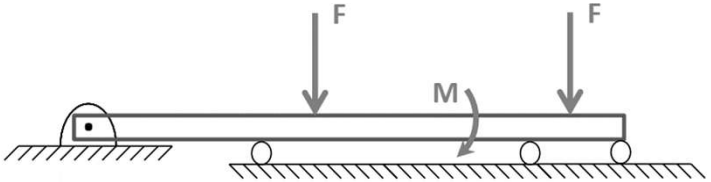
- Beam; a member loaded in a direction perpendicular to its longitudinal axis



Pure Bending in Simply Supported Beam



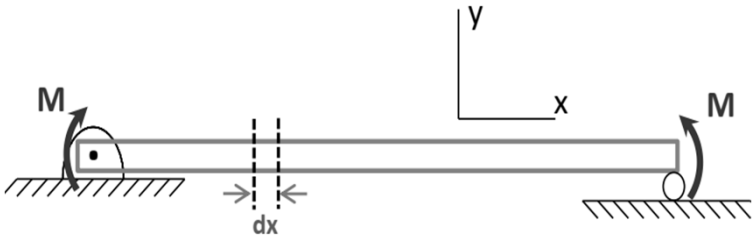
Pure Bending in Cantilever

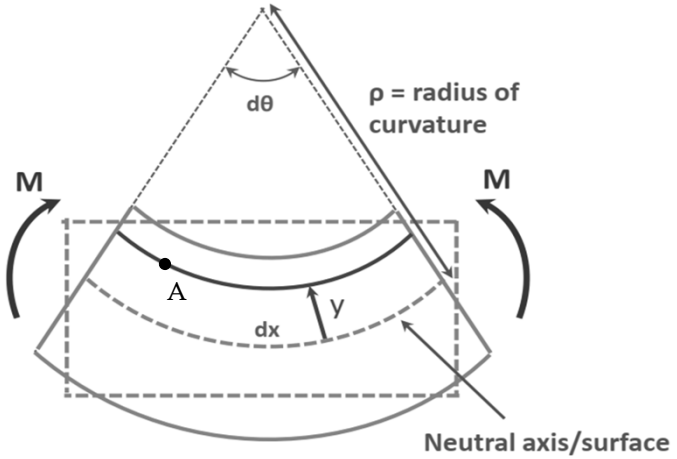


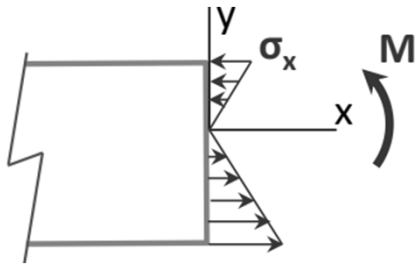
Bending with Shear in Continuous Beam

Elastic Flexural Stress

- Assuming;
  - Pure bending
  - Constant Bending Moment constant

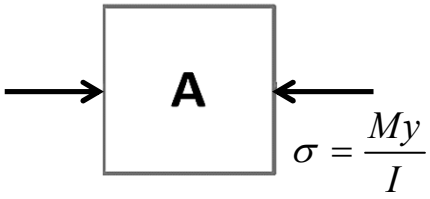






At point A, at a distance *y*, from the neutral axis,

$$\sigma_x = \frac{My}{I}$$

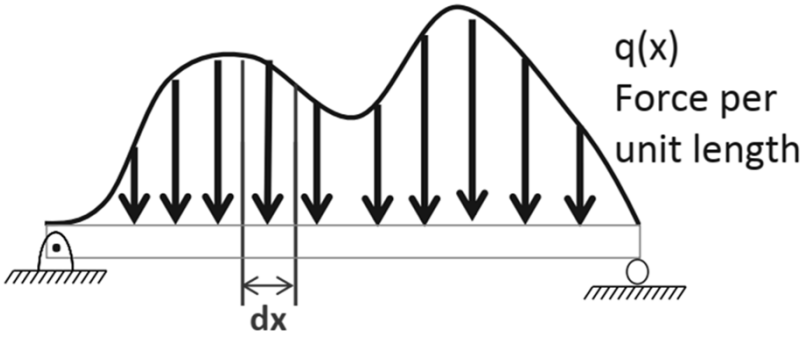


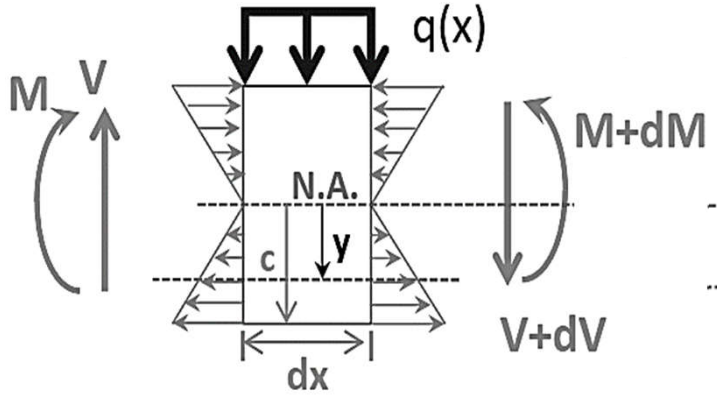
Stress Block/Element

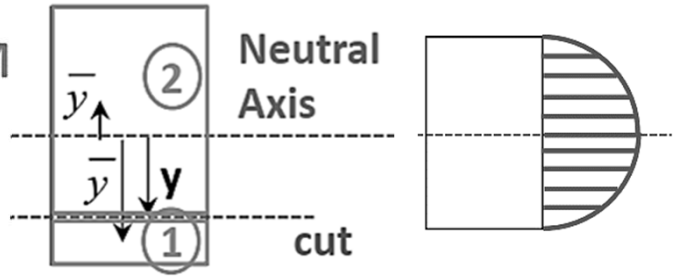
Note: I/y = S, Section Modulus

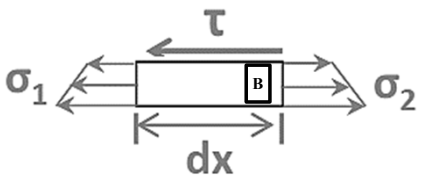
Transverse Shear Stress

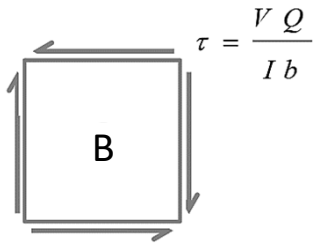
Varying moment along length of beam due to shear forces











Stress Block/Element

$$\tau = \frac{VQ}{Ib}$$

$$Q = \bar{y}A$$

A is the area above (or below) the "cut"

$\bar{y}$ is the distance from the neutral axis to centroid of the area above or below the "cut".

b is the width of the section at the horizontal line where the shear is being calculated.

Stress variation in elastic beams

